

Matukumilli V. D. Prasad

Materials Theory Group
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EMPLOYMENT:

Research Associate Theoretical Sciences Unit, Jawaharlal Nehru Centre for Advanced Scientific Research (JNCASR), Bengaluru.	May 2017 — Present
Research Consultant Advanced Technology Development Centre, Indian Institute of Technology Kharagpur.	Aug 2015 — May 2017
Junior Engineer SYNERGIES CASTINGS Limited, VSEZ, Visakhapatnam, AP.	May 2006 — May 2007

RESEARCH INTEREST:

- *Molecular motors* [[Nano Lett.](#) **16** (2016) 2174]
- *Atomic scale friction* [[Nano Lett.](#) **17** (2017) 2131]
- *Phonon energy dissipation in co-axially sliding CNTs* [[J. Appl. Phys.](#) **118** (2015) 244906]
- *Heat transport in phononic crystals* [[Phys. Rev. B](#) submitted] [[JACS](#) **141** (2019) 20293] [[Springer](#) (2019) 43]
- *Solid-state heat harvest*

EDUCATION:

Indian Institute of Technology Kharagpur, WB • Ph.D. in Nanofriction and Phonon Dynamics. • Advisor: Prof. Baidurya Bhattacharya Thesis Committee Members: Prof. L S Ramachandra, Prof. Nirupam Chakraborti, Prof. Sanjoy Bandyopadhyay . Dissertation Title: "Phonon Scattering Dynamics of Sliding Motion in Carbon Nanotube Oscillators"	2008 — 2015
Andhra University College of Engineering, Visakhapatnam, AP • B.E. in Mechanical Engineering.	2001 — 2005

GRANTS EXPERIENCE:

- **Research Consultant** (ISRO SPONSORED PROJECT, **PI: Baidurya Bhattacharya**; Budget INR 20L, 04/2015 — 03/2017): *“Development of Constitutive Equations for CNT Epoxy Nanocomposites”*
- **Sr. Project Officer** (ASL, DRDO SPONSORED PROJECT, **PI: Baidurya Bhattacharya**; Budget INR 30L, 11/2011 — 08/2012): *“A Multi-scale Investigation of the Strength and Durability of Carbon Nanotube-based Nano-electrode Arrays used as Biological Sensors”*
- **Jr. Project Officer** (DBT SPONSORED PROJECT, **PI: Baidurya Bhattacharya**; Budget INR 20L, 11/2009 — 10/2011): *“A Multi-scale Investigation of the Strength and Durability of Carbon Nanotube-based Nano-electrode Arrays used as Biological Sensors”*
- **Jr. Project Officer** (DMRL SPONSORED PROJECT, **PI: Baidurya Bhattacharya**; Budget INR 10L, 11/2008 — 06/2009): *“Simulation Studies of Mechanical Behavior and Failure of Carbon Nanotubes”*
- **Jr. Project Officer** (IIT KHARAGPUR SPONSORED PROJECT, **PI: Baidurya Bhattacharya**; Budget INR 3L, 05/2007 — 09/2008): *“Multiscale Modeling of Small Scale Interfacial Phenomena in Carbon Nanotube Reinforced Composites”*

- Debattam Sarkar, Subarna Das, Vaishali Taneja, Manisha Samanta, Koushik Jagadish, Animesh Das, Monika Bhakar, **Matukumilli V. D. Prasad**, Suresh Perumal, Goutam Sheet, Dirtha Sanyal, Koushik Pal, N Ravishankar, Umesh V. Waghmare, Kanishka Biswas, "Glassy Thermal Transport Triggers Ultra-High Thermoelectric Performance in GeTe", *Advanced Materials* **37**, 12 (2025)
- Animesh Bhui, **Matukumilli V. D. Prasad**, Shuva Biswas, Abdul Ahad, Anupama Ghata, Divya Rawat, Moinak Dutta, Ajay Soni, Umesh V. Waghmare, Kanishka Biswas, "s-d Coupling-Induced Dynamic Off-Centering of Cu Drives High Thermoelectric Performance in TiCuS", *Journal of the American Chemical Society* **147**, 4 (2025)
- Moinak Dutta, **Matukumilli V. D. Prasad**, Juhi Pandey, Ajay Soni, Umesh V. Waghmare, and Kanishka Biswas, "Local Symmetry Breaking Suppresses Thermal Conductivity in Crystalline Solids", *Angewandte Chemie*. **134**, 15 (2022)
- Moinak Dutta, Shidaling Matteppanavar, **Matukumilli V. D. Prasad**, Juhi Pandey, Avinash Warankar, Pankaj Mandal, Ajay Soni, Umesh V. Waghmare, and Kanishka Biswas, "Ultralow Thermal Conductivity in Chain-like TlSe Due to Inherent Tl⁺ Rattling", *Journal of the American Chemical Society* **141**, 51 (2019)
- **Matukumilli V. D. Prasad** and Baidurya Bhattacharya, "A Phonon Based Understanding of Friction at Small Scales", *Annals of INAE*. **14**, 94 (2017)
- **Matukumilli V. D. Prasad*** and Baidurya Bhattacharya, "Phononic Origins of Friction in Carbon Nanotube Oscillators", *Nano Lett.* **17**, 2131 (2017)
– Media hits: featured story by [Nanowerk](#), [Nanotechweb](#), [IITKGP News](#)
- **Matukumilli V. D. Prasad*** and Baidurya Bhattacharya, "Phonon Scattering Dynamics of Thermophoretic Motion in Carbon Nanotube Oscillators", *Nano Lett.* **16**, 2174 (2016)
- **Matukumilli V. D. Prasad** and Baidurya Bhattacharya, "Phonon Wave-Packet Scattering and Energy Dissipation Dynamics in Carbon Nanotube Oscillators", *J. Appl. Phys.* **118**, 244906 (2015)
- **Matukumilli V. D. Prasad** and Baidurya Bhattacharya, "Buckling and Post-Buckling Characteristics of Cantilevered Single-walled Carbon Nanotubes in Bending", *Int. J. Nanosci.* **14**, 1550019, (2015)
- **Matukumilli V. D. Prasad** and Baidurya Bhattacharya, "Molecular Dynamics Simulations of Carbon Nanotube-based Oscillators having Topological Defects", *Int. J. Nanosci.* **10**, 355 (2011)

INVITED BOOK CHAPTERS

- **Matukumilli V. D. Prasad** and Umesh V. Waghmare, "Theory and simulations of lattice thermal conduction". In: ed. by Paolo Mele, Dario Narducci, Michihiro Ohta, Kanishka Biswas, Juan Morante, Shrikant Saini, Tamio Endo, *Thermoelectric thin films: materials and devices* Springer, Jun 2019.

AWARDS AND HONORS:

- **Outstanding Research Award** 2016-17 (National PARAM Supercomputing Facility, Pune)
— *one among the top five outstanding research works carried-out using PARAM YUVA II in the year 2016-17*
- **Best Poster Presentation Award** Nov 2016 (National Symposium on Ultrasonics, S. N. Bose National Centre for Basic Sciences, Kolkata)

TEACHING EXPERIENCE:

Teaching assistant

December 2016.

- GIAN course on Nonequilibrium Statistical Mechanics and Molecular Dynamics, organized by IIT Kharagpur, Winter 2016

Mentoring student, IIT Kharagpur

2013 — 14.

- Guide Nishanth K S, an undergraduate student of Civil Engineering Department, in using Molecular Dynamics to investigate thermal transport characteristics of metal nanoclusters encapsulated inside carbon nanotube.

PROFESSIONAL ACTIVITIES AND SERVICE:

- Reviewer for *ACS Applied Energy Materials*, *Smart Materials and Structures*, *Nanotechnology*, *Materials Research Express*, *Computational Materials Science* and *Nanoscale*.
- Member of American Physical Society (APS), Indian Society for Heat and Mass Transfer (ISHMT), American Society of Mechanical Engineers (ASME)

INVITED TALKS AND SEMINARS: (* SPEAKER)

- **Matukumilli V. D. Prasad**, and Baidurya Bhattacharya*. "*Mechanical Properties of CNTs and CNT Nanocomposites Through ab-initio and Atomic Simulations*", Dept of Civil Engg, Johns Hopkins University, Baltimore, MD, USA, Oct 2012
- **Matukumilli V. D. Prasad**, and Baidurya Bhattacharya*. "*Molecular Dynamics Simulations of Thermal Transport in Carbon Nanotubes*", Annual Technical Meeting of NMD-ATM, College of Engineering Pune, Pune, India, Nov 2014

- **Matukumilli V. D. Prasad**, and Baidurya Bhattacharya*. "*Phonon Scattering Dynamics of Friction and Thermophoretic Motion in Carbon Nanotube Oscillators*", International Centre for Mathematical Sciences, Edinburgh, UK, Oct 2015
- **Matukumilli V. D. Prasad***. "*Carbon Nanotube Oscillators: Phonons and Friction*", Journal Club Seminar at Physics Department, IIT Kharagpur, India, Oct 2016
- **Matukumilli V. D. Prasad***, and Baidurya Bhattacharya. "*Molecular dynamics of thermal-phonon induced drive: Coherent and diffusive scattering regimes in nanomechanical devices*", 4th International Conference on Advances in Materials & Materials Processing, IIT Kharagpur, India, Nov 2016
- **Matukumilli V. D. Prasad***, and Umesh V. Waghmare. "*Phonon drag effect for thermophoretic and thermoelectric energy conversion applications*", Frontiers of Mechanics colloquium, IIT Kharagpur, India, 24-26 Nov 2017
- **Matukumilli V. D. Prasad**, and Umesh V. Waghmare*. "*Anharmonic phonon thermal transport*", India-UK workshop on Thermoelectric Materials for Waste-Heat Harvesting, JNCASR, India, Jan 2018
- **Matukumilli V. D. Prasad***, and Baidurya Bhattacharya, "*Development of Constitutive Equations for CNT-Epoxy Nanocomposites*", Project Closure Presentation, Vikram Sarabhai Space Center (VSSC), Thiruvananthapuram, India, Feb 2018

CONFERENCE ORAL/POSTER PRESENTATIONS: (* SPEAKER)

- **Matukumilli V. D. Prasad***, and Baidurya Bhattacharya. "*Molecular Dynamics Simulations of Ni Nanowire Bending*", Molecules and Materials – New Directions, Jawaharlal Nehru Centre for Advanced Scientific Research, Bengaluru, India, Dec 2008
- **Matukumilli V. D. Prasad***, and Baidurya Bhattacharya. "*Bending Properties of Carbon Nanotubes with Multiple Vacancy Defects using Atomistic Simulations*", AIMTDR 2008, 2nd International 23rd All India Manufacturing Technology, Design and Research Conference, IIT Madras, India, Dec 2008
- **Matukumilli V. D. Prasad***, and Baidurya Bhattacharya. "*Molecular Dynamics Simulations of Carbon Nanotube-based Oscillators having Topological Defects*", ICANN 2011, International Conference on Advanced Nanomaterials and Nanotechnology, IIT Guwahati, India, Nov 2011
- **Matukumilli V. D. Prasad***, and Baidurya Bhattacharya. "*Energy Dissipation and Phononic Friction in Carbon Nanotube Oscillators*", ICTACEM 2014, 6th International Conference on Theoretical, Applied, Computational and Experimental Mechanics, IIT Kharagpur, India, Dec 2014
- **Matukumilli V. D. Prasad***, and Baidurya Bhattacharya. "*Carbon Nanotube Oscillators: Effect of Small Bending Strain*", ICTACEM 2014, 6th International Conference on Theoretical, Applied, Computational and Experimental Mechanics, IIT Kharagpur, India, Dec 2014

- **Matukumilli V. D. Prasad***, and Baidurya Bhattacharya. "*Acoustic Phonon Propagation and Interfacial Scattering in Nanostructures*", NSU 2016, National Symposium on Ultrasonics, S. N. Bose National Centre for Basic Sciences, Kolkata, India, Nov 2016 **Awarded best Poster Presentation/Prize**
- **Matukumilli V. D. Prasad**. Attended the QPC 2017, Conference on Quantum Physics and Consciousness, Bhaktivedanta Institute & Bharatiya Vidya Mandir, Kolkata, India, Feb 2017
- **Matukumilli V. D. Prasad***, Koushik Pal, and Umesh V. Waghmare. "*Phonon thermal transport in ternary Zintl compounds: Bi_2Mg_2X ($X = Mg, Ca$)*", India-UK workshop on Thermoelectric Materials for Waste-Heat Harvesting, JNCASR, India, Jan 2018

WORKSHOPS/SCHOOLS:

- "Mechanics over Micro and Nano Scales (STCMMN 09)", Bengal Engineering and Science University, Shibpur, Kolkata, Dec 2009
- "Electronic Structure and Properties of Low Dimensional systems", Physics Department, IIT Kharagpur, Apr 2016
- "Lithium Ion Battery Technology and Mathematical Modeling", Metallurgical and Materials Engineering Department, IIT Kharagpur, May 2016
- "Winter School-2017 on Frontiers in Materials Science", JNCASR, Dec 2017
- "India-UK workshop on Thermoelectric Materials for Waste-Heat Harvesting", New Chemistry Unit, JNCASR, Jan 2018
- "Sorbonne-JNCASR School on Advanced Computational Materials Science (SJSACMS-2018)", Thematic Unit of Excellence in Computational Materials Science, JNCASR, Feb 2018
- "Young Scientist Discussion Meeting on Chemistry of Nanomaterials", New Chemistry Unit, JNCASR, Jul 2018
- Winter School on "Modelling and Simulations of Materials for Energy and Environment", Theoretical Sciences Unit, JNCASR, Dec 2018
- "Ten years of ICMS celebrations and Annual SAMat symposium", International Center for Materials Science, JNCASR, Jan 2019
- "Modern Approaches in Chemistry & Biology-2020 (MACB-2020)" as a celebration of the 80th birthday of Professor Mike Klein, JNCASR, Feb 2020
- "Moving ions with VASP" Online VASP workshop, September 27-30, 2022
- "New Approaches and Machine learning Methods for Ab initio calculations: NAMMA Psi-k Workshop 2023" IISc and JNCASR, July 2023

COMPUTATIONAL SKILLS:

- **First-Principles and Atomistic Simulations:** Extensive experience with VASP, QUANTUM ESPRESSO, and classical molecular dynamics packages including LAMMPS, GROMACS, GULP, and MEDEA; complemented by self-developed MD and MC codes for system-specific simulations.
- **Phonon and Thermal Transport Analysis:** Proficient in PHONOPY, PHONO3PY, D3Q, THERMAL2,

PYTHON-SSCHA, PHONTS, SHENGBTE, ALMABTE, and ALAMODE.

- **Programming & Scripting:** Fortran, C, C++, and MPI parallel programming; experienced in Python-based workflows for high-throughput and data analysis.
- **Mathematical and Data Analysis Tools:** MATLAB/Octave, Mathematica, GNUplot, Matplotlib, and xmgrace.
- **Structure and Trajectory Visualization:** ATOMEYE, VMD, V-SIM, Jmol, XCRYSDEN, OVITO, VESTA, and ATOMSK.
- **Operating Systems:** Advanced user of Linux-based environments; experienced with Windows.